

MATH0014 Algebra 3: Further Linear Algebra

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0014
<i>Level:</i>	5 (UG)
<i>Normal student group(s):</i>	UG: Y2 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	85% examination, 15% coursework
<i>Normal Pre-requisites:</i>	MATH0006 (Algebra 2)
<i>Lecturer:</i>	Dr R Reynolds

Course Description and Objectives

This module builds on the linear algebra developed in Algebra 1 and Algebra 2, going deeper into the subject and presenting more advanced constructions and results. One of the aims is to take the ideas and techniques we learned for ordinary vectors and matrices and become comfortable applying them to more abstract vector spaces and linear maps, including to infinite-dimensional cases. Another theme is diagonalisation and its generalisations; this is the process of choosing a good basis in which to study a linear map.

The final section of the course studies inner products—the abstraction of the familiar dot product—and more general quadratic and bilinear forms. This includes the general theory as well as practical methods for working with them.

Recommended Texts

- S Axler, *Linear algebra done right* (Springer)

Detailed Syllabus

- Revision and extension of material from Algebra 1: vector spaces, basis, dimension, subspaces, linear maps, kernel and image, isomorphism.
- Direct sums: criterion for a vector space to be a direct sum, projections, complementary subspaces.
- Determinants and diagonalisation: determinants, eigenvectors, eigenspaces, diagonalisation, triangularisability, the Cayley–Hamilton theorem over $\mathbb{C}$, Jordan normal form (statement and examples).
- Dual spaces: dual bases, dual of a linear map.
- Inner products and spectral theorems: examples in infinite dimensions including integrals on function spaces, adjoints, Hermitian inner products, orthogonal complements, self-adjointness, normal matrices and diagonalisability.
- Bilinear forms: symmetric and quadratic forms, signature, tensor products.